

Integrating LLM, Semantic Perception and Spatial Reasoning for Improved Robot Action Control

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Abstract. Despite recent progress in integrating Large Language Models (LLMs) to enhance robotic capabilities, key challenges remain—particularly in interpreting perceptions, interacting with humans, and executing tasks in real-world environments. These challenges are largely due to LLM hallucinations and lack of formal semantic knowledge that hinder contextual understanding. To address this issue, we propose a semantic framework that combines an LLM and a commonsense knowledge graph to align the robot’s perceptions with a suitable task execution plan. The framework integrates pre-trained deep learning models for object detection, face identification, speech recognition, an LLM, and a local knowledge graph based on Resource Description Framework (RDF). The knowledge graph including scene and object description, as well spatial knowledge is used to select and execute the suitable task plan for the given context. Experimental results demonstrate the effectiveness of framework’s components in improving the robot’s interaction and and actuation in its environment.

1 Introduction

Recent progress in Large Language Models (LLMs) has transformed artificial intelligence, offering new opportunities to enhance the cognitive abilities of humanoid robots in everyday tasks and human assistance. Yet, significant challenges persist—particularly in perception interpretation, human interaction, and real-world task execution—mainly due to LLM hallucinations and the absence of structured semantic knowledge, which limit contextual understanding. This paper introduces a semantic framework that addresses these limitations in the Pepper humanoid robot by integrating an RDF knowledge graph with Gemini 1.5 Flash, an advanced LLM that facilitates natural language interaction and coordinates specialized perception modules. The goal is to enrich Pepper’s cognitive functions—such as contextual reasoning, object and face recognition, and spatial awareness—to support autonomous, goal-driven behaviors like object manipulation.

The architecture of the framework integrates several pre-trained models, each specialized for a specific perceptual task: YOLOv8n for real-time object detection and classification, DLIB for facial recognition and identity memory, and MiDaS for monocular depth estimation, providing 3D spatial awareness. The output from MiDaS is

further processed through a mathematical framework involving coordinate transformations and inverse kinematics, enabling precise spatial reasoning and control. Additionally, a domain-specific ontology supports querying the RDF knowledge graph to extract high-level semantic knowledge about objects, their locations, and the appropriate action plans that the robot can use to execute tasks based on vocal instructions extracted by the LLM. A validation experiment is presented in [this demo](#)¹ demonstrate the effectiveness of the proposed approach.

2 Related work

Recent advances in autonomous robotics focus on combining vision and language models to improve tasks execution, question answering and environment understanding. SayCan [1] combines LLMs with affordance-based value functions to translate high-level instructions into executable actions. While the LLM handles reasoning in natural language, an evaluation function assesses action feasibility. Actions are selected based on both semantic relevance and physical affordance. However, the approach still depends on external modules for perception. PIVOT [5] approaches robotics as a Visual Question Answering (VQA) task by annotating the environment with possible actions and querying a VLM to select the optimal move. While it avoids explicit training, the method is limited by depth ambiguities and lacks an adaptive correction mechanism. VILA [4] proposes a unified approach to long-term robotic planning by directly using Vision-Language Models (VLMs) to generate actions from visual inputs and textual instructions, without external modules. This enables multimodal reasoning, improved spatial and object understanding, and closed-loop visual feedback, outperforming traditional planners in both real and simulated settings. It reflects the broader trend of combining pre-trained visual encoders with LLMs for tasks like VQA, captioning, and Optical Character Recognition (OCR). [6] provide useful representations for visuomotor tasks but remain limited in their ability to generalize robotic control. In [2], ELMiRA is proposed to demonstrate modular and integrated approach to enhance human-robot interaction by combining speech, vision, and control modules for flexible, language-driven execution. [3] introduces OWL×T5, a model that combines OWLv2 and CrossT5 to enable zero-shot recognition and manipulation of unseen objects, while

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¹ <https://lissi.fr/lissi2/prototypes-et-demonstrateurs/demo1-2025>

maintaining LLM-level language understanding. Tested in simulation, it demonstrates the effectiveness of integrated vision-language-action models for adaptive robotic interaction in open environments.

However, despite the contribution of these approaches, they lack formal knowledge representation mechanisms to enrich model inference with semantic context and reduce hallucinations when converting LLM outputs into actionable knowledge.

3 Framework Architecture

The framework architecture consists of the following components: Contextual dialogue manager, Object detection and recognition, Face recognition and person identification, Orchestrator, Knowledge Graph and Object grasping controller composed of a depth estimator and solver to generate the action to be executed by the robot, see figure 1.

3.1 Contextual dialogue manager

To enable the Pepper robot managing naturally contextual dialogues with multi-domain responsiveness, we integrated Microsoft's Gemini 1.5 Flash large language model into a modular speech-processing pipeline. Voice queries are transcribed to text, processed by the LLM, and vocalized via Pepper's speech synthesis system using SoftBank's Naoqi Python library. While this setup allows Pepper to autonomously generate context-aware responses, it initially lacked the fluidity of human conversation, particularly the ability to manage interruptions or corrections in real time. To address this, we implemented two dynamic dialogue control mechanisms: the Direct Stop Event, which detects predefined interruption keywords like "stop" or "that's not right," and the Fluent Stop Event, which estimates the robot's speaking duration to allow real-time response overrides. Together, these enhancements significantly improve the naturalness and interactivity of human-robot dialogue.

3.2 Object detection and recognition

To enable visual scene understanding, we integrated YOLOv8n, a lightweight object detection model, with Microsoft's Gemini 1.5 Flash LLM and SoftBank's Naoqi framework. Upon receiving a user query (e.g., "What is on the table?"), the system activates Pepper's camera, captures an image, detects and labels objects via YOLOv8n, and sends those labels to Gemini 1.5 Flash to generate a structured scene description. This output is then vocalized by Pepper. The choice of YOLOv8n was based on its superior real-time performance compared to alternatives like Faster R-CNN and SSD, balancing speed, accuracy, and hardware efficiency. Furthermore, YOLOv8n supports environment-specific retraining, enabling adaptable perception in dynamic contexts such as industrial settings. This modular visual pipeline enhances Pepper's contextual awareness and extends its autonomous interaction capabilities.

3.3 Face recognition and person identification

To enable Pepper to autonomously recognize new individuals and identify them based on their facial features, we implemented an incremental learning pipeline using dlib. This pipeline incorporates a ResNet-34 convolutional neural network for face encoding and a Support Vector Machine (SVM) for classification. However, the feature extraction step relies on the Histogram of Oriented Gradients (HOG) method.

In the context of face encoding, each newly detected face is transformed into an embedding vector $e \in \mathbb{R}^n$, typically with $n = 128$. To determine whether the person has already been encountered, the robot compares this new vector to those stored in its memory, which represent previously seen individuals. This comparison is based on the *Euclidean distance* $d_{\text{euclidean}}$ between embedding vectors. A match is considered to exist if the following condition is satisfied:

$$d_{\text{euclidean}} < 0.60$$

Dlib was selected over alternative models such as FaceNet and OpenCV thanks to its optimal trade-off between accuracy, computational cost, and integration feasibility. Performance evaluation under two scenarios—Variable Distance and Visual Occlusion—revealed robust accuracy up to 3 meters and resilience to minor occlusions (e.g., blur, sunglasses), though full-face masks significantly impaired recognition accuracy as shown on Figure 2. These results confirm the pipeline's effectiveness for real-time human-robot interaction in semi-controlled environments.

3.4 Orchestrator and Semantic Knowledge Graph

The knowledge graph contains RDF-based factual knowledge about the various indoor regions where the robot is permitted to move, as well as information about objects, their locations, and the possible plans the robot can follow to perform actions on these objects according to human instructions extracted by the LLM. The OWL ontology defines general concepts such as Object, Action, Plan, and Indoor Region, along with the relationships between them, such as `isLocatedIn`, `enablesAction`, and `suitablePlanForAction`. The Orchestrator enables SPARQL queries on the knowledge graph to retrieve relevant information about the plan followed by Midas and the actions to be executed by the robot. For example, a query can retrieve the coordinates of an action involving the movement of an object—an instance of the throwable class—into a trash bin.

3.5 Object grasping controller

To enable Pepper's precise 3D perception and human-like interaction with objects, we implemented a dual scenario approach. When a user requests an object on a table, Pepper either retrieves it directly if explicitly specified, using its RDF knowledge graph to localize the *table* instance with known coordinates, or handles vague requests by first moving to the table, capturing an image, and applying the YOLO model for object recognition. The detected object is then matched with RDF instances to disambiguate the user's description.

Depth estimation is then performed using the MiDaS model, which produces a depth map converting 2D pixel coordinates into 3D points relative to Pepper's camera. MiDaS was chosen over DepthNet and Monodepth2 due to its superior accuracy and robustness in varying perspectives and lighting, despite higher computational requirements.

The core equations underlying object detection using the depth map generated by MiDaS is given below.

- **Image Coordinate Representation:** A 3D scene captured by a camera is projected onto a 2D image plane, where each real-world point corresponds to a pixel (u, v) . This position is expressed in homogeneous coordinates to facilitate transformation and depth reasoning.

$$p_{\text{image}} = \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \quad (1)$$

with:

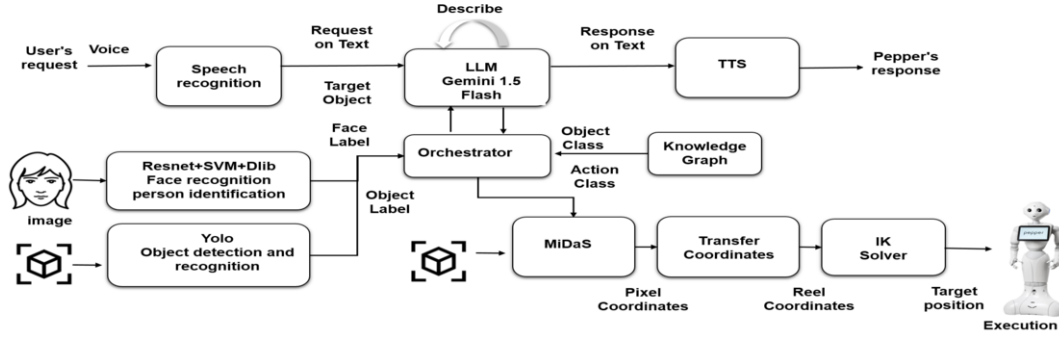


Figure 1. The proposed Architecture for Semantic Scene Understanding and Robotic Execution with Pepper robot a scaling factor S :

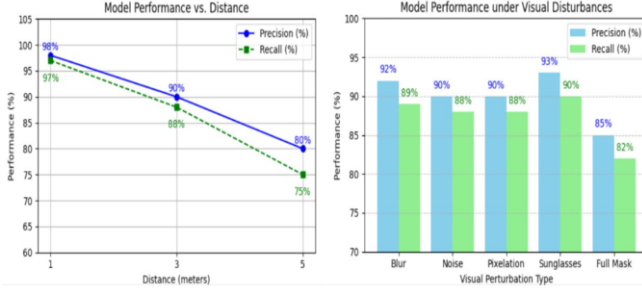


Figure 2. Facial Recognition Performance under Distance and Occlusion Variations

- u, v : coordinates of the point in the image (in pixels),
- 1: added to convert to homogeneous coordinates, facilitating matrix calculations.
- **Camera intrinsic matrix**: The intrinsic parameters of the camera define how it projects 3D points into 2D. These parameters are grouped in the calibration matrix K :

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

where:

- f_x, f_y : focal lengths in pixels,
- c_x, c_y : coordinates of the optical center of the image.

To retrieve relative 3D coordinates from image coordinates, we use the inverse of K :

$$K^{-1} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}^{-1} \quad (3)$$

- **Conversion from 2D to 3D**: Once K^{-1} is obtained, we can convert the image coordinates into a directional vector in the camera space:

$$p_{\text{camera}} = K^{-1} \cdot p_{\text{image}} = \begin{bmatrix} X' \\ Y' \\ 1 \end{bmatrix} \quad (4)$$

- **Depth integration**: The normalization of coordinates is not sufficient to recover the real 3D position. We need a depth map Z_d which is the output prediction of the MiDaS model:

$$P_{3D \text{ relative}} = Z_d \cdot \begin{bmatrix} X' \\ Y' \\ 1 \end{bmatrix} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \quad (5)$$

with:

$$X = Z_d \cdot X', \quad Y = Z_d \cdot Y', \quad Z = Z_d \quad (6)$$

- **Conversion into meters**: The depth values obtained by prediction models are generally not in real-world units. Therefore, we apply

$$S = \frac{Z_{\text{real reference}}}{Z_{\text{estimated reference}}} \quad (7)$$

This factor is then applied to the relative 3D coordinates:

$$P_{3D \text{ meters}} = S \cdot P_{3D \text{ relative}} \quad (8)$$

The 3D coordinates from the camera frame are transformed into Pepper's movement frame via a coordinate conversion:

$$\begin{cases} X' = Z \\ Y' = -X \\ Z' = L - Y' \end{cases} \quad (9)$$

where L is the camera offset from Pepper's center of gravity.

Finally, Pepper applies inverse kinematics to position its arm for grasping. Using a two-link planar arm model, joint angles θ_1 (shoulder) and θ_2 (elbow) are computed from the target (X', Z') via:

$$\theta_2 = \arccos\left(\frac{X'^2 + Z'^2 - L_1^2 - L_2^2}{2L_1L_2}\right) \quad (10)$$

$$\theta_1 = \arctan 2(Z', X') - \arctan 2(L_2 \sin \theta_2, L_1 + L_2 \cos \theta_2) \quad (11)$$

This pipeline enables real-time, accurate object retrieval and manipulation in natural interaction scenarios.

4 Conclusion

This paper presents a flexible cognitive architecture designed to enhance human-robot interaction, addressing the cognitive limitations of humanoid robots like Pepper in dynamic and unstructured environments. This architecture replaces centralized, service-dependent approaches with distributed cognitive modules coordinated by Gemini 1.5 Flash. It integrates: LLMs for language understanding and control, vision models (YOLOv8n, DLIB, MiDaS) for object detection, face recognition, and depth estimation, and a domain-specific knowledge graph for semantic reasoning and command disambiguation. By combining multimodal perception with symbolic reasoning, the proposed architecture enhances Pepper's situational intelligence, enabling context-aware interaction, autonomous decision-making, and scalable deployment in real-world settings.

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